

Activity 2

Aim: To determine pH of different fruit juices by using universal indicator or pH paper.

Apparatus: Test tubes, droppers, glass rod, beakers, etc.

Chemicals: Different fruit juices, pH paper or universal indicator.

Procedure:

1. Take 5 ml of given different fruit juices in separate A, B, C, D marked test tubes.
2. Put pH paper or add 2 drops of universal indicator in each test tube and shake it well.
3. Observe the color of pH paper or solution developed in each test tube.
4. Determine the pH of each solution by comparing the colors with the standard pH color strips.
5. Find out nature of given fruit juices on the basis of pH values.

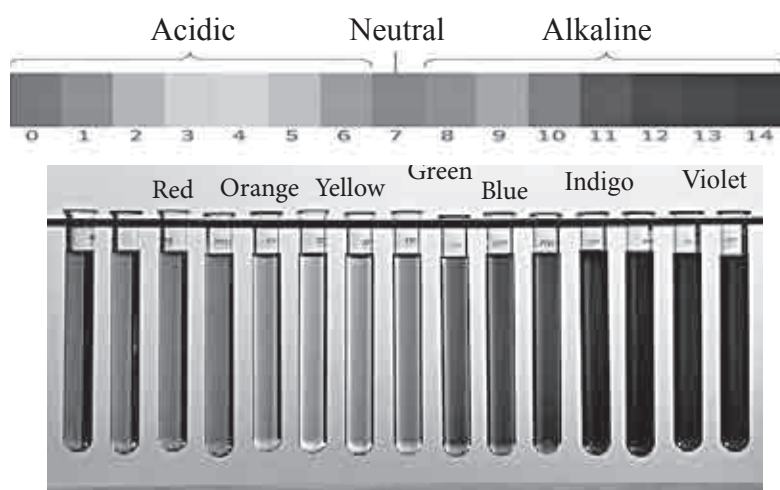


Fig. Activity 2.1 pH scale

Observation table and Conclusion :

Sr. No.	Juice sample	Color of pH paper or solution of universal indicator	pH	Nature
1	Lemon...(A)	Red	2-3	acidic
2	Tomato...(B)	Light Red	4-5	acidic
3	Orange...(C)	Orange	3-4	acidic
4	Beet...(D)	Purple	5-6	Base

Remark and sign of teacher:

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. In a 5 mL solution of lemon juice add 2 drops of universal indicator, the color of solution becomes red-orange, the pH of solution is
a. ☒ > 7 c. equal to 7
b. ☐ < 7 d. between 7 to 14
2. The pH of grape juice is about 3.3, the nature of grape juice is
a. ☒ Acidic c. neutral
b. alkaline d. acidic as well as alkaline
3. When a strip of pH paper is dipped in a solution of orange juice, the color of pH paper become orange the pH of solution is
a. ☒ > 7 c. equal to 7
b. ☐ < 7 d. between 7 to 14

Short answer questions

1. What is the pH of pure water?

Ans. The pH of pure water is 7.

2. On mixing any two fruit juices, would pH remains same or differ? Verify your answer experimentally.

Ans. The pH of the mixture may differ based on the acidity or alkalinity of the fruit juices being mixed.

3. What is the pH range of universal indicator?

Ans. pH range of universal indicator generally is between 4 to 14. But some universal indicators with range 1 to 14 are also available.

Remark and sign of teacher: